Spring 2026 Capstone Assignment 2

Analyzing a Neo4j Graph Database via Cypher + GDS

***** You may work in small groups of no more than three (3) people for this Assignment *****

For our second Capstone assignment, we will be using both **Cypher** and **Neo4j's Graph Data Science (GDS) library** to analyze the content of a graph database comprised of data provided by the US Government's Food and Drug Administration (**FDA**). Specifically, the graph has been created using information available within the FDA's *Adverse Event Reporting System* (**FAERS**) https://www.fda.gov/drugs/surveillance/fdas-adverse-event-reporting-system-faers. FAERS is a computerized information database designed to support the U.S. Food and Drug Administration's (FDA) post marketing safety surveillance program for all approved drug and therapeutic biologic products. For purposes of this assignment, we will refer to any drug or therapeutic biologic using the generic term "drug".

The FDA uses FAERS to track new adverse events and medication errors that might occur when these products are administered to patients. It is a system that measures occasional harms from medications to ascertain whether the risk–benefit ratio is high enough to justify continued use of any drug and to identify correctable and preventable problems in health care delivery (such as need for retraining to prevent prescribing errors).

Reporting of adverse events from the point of care is voluntary in the United States. The FDA receives some adverse event and medication error reports directly from health care professionals (such as physicians, pharmacists, nurses, and others) and consumers (such as patients, family members, lawyers, and others). Health professionals and consumers may also report these events to the products' manufacturers. If a manufacturer receives an adverse event report, they are required to send the report to the FDA as specified by regulations. See this web page for **FAERS** Frequently Asked Questions: https://fis.fda.gov/extensions/FPD-FAQ/FPD-FAQ.html#_Toc514144622 .

Loading the FAERS Database within Your Neo4j Environment

A Neo4j 'dump' file containing the pre-built graph database (**healthcare-analytics-50.dump**) is provided via the Canvas page for this assignment. Download a copy of that file to your local computing environment.

To get started with the graph database, you will first need to install **version 2.1.1** of **Neo4j's Desktop software**: https://neo4j.com/deployment-center/?desktop-gdb

Next, you will need to create a new **INSTANCE** having the name "**Capstone**" via the Neo4j Desktop 2.1.1 user interface. Once the new instance is created, activate it by clicking the "**Start Instance**" button (appears as a rightward-pointing triangle within a circle).

Next, create a new Neo4j database within the "Capstone" instance using the name "**Healthcare**". If successful, the new Database should appear within the "**Databases**" drop-down menu shown within the "Capstone" instance interface.

Next, temporarily **STOP** the Capstone instance by clicking on the "**Stop Instance**" button. While the Capstone instance is stopped, ***install the GDS, GenAI, and APOC "plug ins"*** within the instance by selecting the "**More Instance Actions**" dropdown menu (the **three horizontal dots** shown to the far right of the "Capstone" instance name). Within that dropdown, click the "**Plugins**" item and then click the individual "**Install**" buttons for **APOC**, **GenAI**, and **Graph Data Science**.

Next, return to the "**Databases**" dropdown menu and click on the "**Open Menu**" option (the **three horizontal dots** on the righthand side of the screen) for the **Healthcare** database. Select the "**Load Database from File**" option and select the previously downloaded **healthcare-analytics-50.dump** file. Once the loading of the database has been completed, the "**Healthcare**" database is ready for your use.

FAERS Data Description

As mentioned above, FAERS is used by the FDA to track adverse events and medication errors that might occur when drugs are administered to patients. The FAERS graph database is comprised of eight (8) types of nodes having the following Labels:

Drug: Drug that potentially contributed to the adverse event. A drug can be one of four suspect types**: a primary suspect, secondary suspect, concomitant** or **interacting drug responsible for the adverse effect**. The specific suspect type is identified by the relationship between Case and Drug Nodes

Case: Demographic information of a person/patient who experienced the adverse event

Reaction: The specific reaction that the patient ("Case") developed after consumption of the respective drug, e.g., 'Pain', 'Body temperature increased' or 'Insomnia' or 'Memory Loss' etc. There are a large number of distinct reactions that are possible across the population of drugs represented within FAERS.

ReportSource: Who was responsible for reporting the adverse event to FAERS, e.g., 'Health Professional', 'Consumer', 'User Health Facility' , etc.

Outcome: The long term outcome of the case after the occurrence of the adverse event, e.g., 'Hospitalization: Initial or Prolonged', 'Disability', 'Death', etc.

Therapy: If the patient is receiving the drug as a part of a proscribed therapy, a description of that therapy is provided here.

Manufacturer: The name of the pharmaceutical company that manufactures the drug

AgeGroup: Demographic indicator identifying the age group to which the patient belongs.

Each of the types of nodes has varying characteristics, e.g., each has different properties; certain types of relationships originate at certain types of nodes and are directed at other specific types of nodes, etc. Furthermore, there are many different types of relationships that exists within the knowledge graph.

As the Data Science practitioner, you will be responsible for discovering and documenting many of these graph-specific characteristics as part of your work for this assignment.

Analyzing the FAERS Graph Database

Since an explanation of the entire content of the FAERS graph is not readily available, you will first be using your Cypher skills to perform some specific EDA tasks for purposes of helping you clarify your understanding of the content of the graph. Then, after having deepened your domain knowledge, you will be tasked with performing several types of analyses that can potentially be of use to healthcare practitioners responsible for defining and administering drug treatments. The first few pieces of analysis will be performed using advanced Cypher queries, while the remaining items will require the additional use of Neo4j's GDS library. The specific tasks you are required to implement for this assignment are explained below.

Your deliverable for this Assignment is a Jupyter Notebook. It should contain a combination of Cypher commands/queries, the output of the Cypher commands/queries, and explanatory narratives contained within **properly formatted Markdown cells**. The Cypher commands used should be placed within code cells within the Notebook. The output of the Cypher commands/queries can be incorporated via cut/paste into notebook cells.

The EDA-oriented graph database queries are as follows (20 pts):

1) **(2 pts)**: Write and execute a Cypher query that counts the total number of nodes within the Healthcare graph
2) **(2 pts)**: Write and execute a Cypher query that lists all distinct graph labels contained within the Healthcare graph.
3) **(2 pts)** Write and execute a Cypher query that counts the total number of relationships within the Healthcare graph
4) **(2 pts)** Write and execute a Cypher query that lists all distinct types of relationships within the Healthcare graph
5) **(3 pts)** Write and execute a Cypher query that lists all unique properties for each type of node (node type = distinct label)
6) **(3 pts)** Write and execute a Cypher query that lists the distinct genders found within Case node 'gender' property values.
7) **(3 pts)** Write and execute a Cypher query that calculates the number of distinct reactions found within the graph. Note that you will need to determine the appropriate node property to analyze for purposes of answering this question correctly.
8) **(3 pts)** Write and execute a Cypher query that lists all distinct Outcome description values

Deeper Analytical Questions (40 pts)

Now that you are much more familiar with the content of the graph, answer the following analytical questions via Cypher queries. For each question, provide your Cypher query, a written explanation of the query logic, and commentary that explains any conclusions you are able to derive from the results returned by the query.

1) **(10 pts)** Which reactions appear in the graph most frequently? Present your results in descending order, and list only the top 20 reactions.
2) **(10 pts)** Which drugs produced the most severe outcomes? For purposes of answering this question, we'll assume that the four most severe outcomes are as follows: 'Death', 'Life-Threatening', 'Disability', 'Hospitalization'. Present your results in descending order, and list only the top 10 drugs.
3) **(10 pts)** Which manufacturing companies have the largest number of drugs with reported side effects? Present your results In descending order, and list only the top 10 results.
4) **(10 pts)** Referring to the results of the previous question: For the manufacturer having the largest number of drugs with reported side effects, what are the top 5 registered drugs for that company and the side effects from each of those drugs? NOTE: be sure to list only distinct side effects for each drug in your answer.

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Questions Requiring Use of GDS (40 pts)

For each of these questions, you should first create an appropriate in-memory graph projection that contains the graph content that you believe is necessary for applying the GDS algorithm you intend to use. Within your Jupyter Notebook, provide the Cypher code used to create the graph projection, a written explanation of the logic being used to create the projection, the Cypher/GDS query used to apply the GDS algorithm, and commentary that explains any conclusions you are able to derive from the results returned by GDS algorithm.

1. **(20 pts)** Use a node similarity algorithm to Identify "similar patient journeys" where different patients experienced identical sequences of reactions after taking the same drugs. This analysis can help to pinpoint common risk factors amongst patients. It is up to you as the data scientist to select the node similarity algorithm that you believe is most appropriate for this analysis. Be sure to provide a written explanation for your choice of a specific algorithm, and be sure to incorporate the specific context of the question that you have been asked to answer as part of your explanation.

2. **(20 pts)** Use a community detection algorithm to cluster patients based on shared demographic characteristics and shared adverse reactions, with the objective being to identify sub-phenotypes of patients who may be at higher risk for specific drug-reaction combinations, i.e., do patients sharing certain demographic characteristics appear to be more likely to suffer from certain adverse reactions? It is up to you as the data scientist to select the community detection algorithm that you believe is most appropriate for this analysis. Be sure to provide a written explanation for your choice of a specific algorithm, and be sure to incorporate the specific context of the question that you have been asked to answer as part of your explanation.

Your Jupyter Notebook deliverable should be similar to that of a publication-quality / professional caliber document and should include clearly labeled graphics, high-quality formatting, clearly defined section and sub-section headers, and be free of spelling and grammar errors.

Upload / submit your Jupyter Notebook within the provided Canvas submission portal. Be sure to save your Notebook using the following nomenclature: **first initial_last name_C2_assn**" (e.g., J_Smith_C2_assn). ***Small groups should identity all group members at the start of the Jupyter Notebook and each team member should submit their own copy of the team's work within Canvas.***